

Table of contents

Potential: “intrinsic energy of inertia”	3
Force caused by Potential: Electro-Magnetic Field	3
Magnetic Field in Vacuum	3
Magnetic Field in Matter	4
Electric Field in Vacuum	4
Electric Field in Matter	4
Abbreviations	4
Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions) . .	5
Potential derivation	5
elec. Potential structure	6
magn. Potential structure	6
Electric endogen energy	6
Electric induced energy (exogenous excitation)	6
Power:	7
Frequency:	7
Phase:	7
Phaseshift:	7
Refraction:	7
Dispersion:	8
Dielectricity:	8
Permeability:	8
Conductivity:	8
Conductor (charge conductive element for energy transfer) condition:	8
Slow changing current (quasi-stationary, low frequencies) condition:	8
Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):	8
Low Frequency	9
High Frequency	9
Continuity at Rest: Electric Charge	9
Continuity at Movement: Magnetic Field	9
Continuity at Acceleration: Radiation Energy	9
Continuity and periodic (harmonious oscillation) Movement in VACUUM	9
Continuity and periodic (harmonious oscillation) Movement in MATTER	10
Near Field	10
Far Field	10
Dissipation in Vacuum	10
Dissipation in Matter	11
Energy gain by excitation	11

Electromagnetic Energy Exchange in Vacuum	12
Energy Balance	13
Energy Density Flux: Poynting-Vector	14
Local Force Balance	14
Tension Force: Stress-Tensor	15
Momentum Balance	15
Abbreviations:	16
Electromagnetic Energy Exchange in Matter	17
Mechanical Energy Exchange in Vacuum	18
Mechanical Energy Exchange in Matter	18
Thermal Energy Exchange in Vacuum	18
Thermal Energy Exchange in Matter	18
Gravitational Energy Exchange in Vacuum	19
Gravitational Energy Exchange in Matter	19
Localization and Fields of Energy Transport	19
Electromagnetic Energy Exchange in Matter	19
Mechanical Energy Exchange in Vacuum	20
Mechanical Energy Exchange in Matter	20
Thermal Energy Exchange in Vacuum	20
Thermal Energy Exchange in Matter	20
Gravitational Energy Exchange in Vacuum	21
Gravitational Energy Exchange in Matter	21
Localization and Fields of Energy Transport	21
Frequency Cause-Effect	21
Frequency (commonness, probability, likelihood)	21
Frequency contextual Causes	22
Conditions (skin, interface, border, surface)	24
Boundary Conditions	24
Interior Conditions	25

Abbreviations	25
Distinguishing Situations Conditions and Dependencies	25
Spacetime Embedding	25
Movement Framework	25
Medium Property	26
Field Distance	26
Localisation Identity	26
Operators	26

Potential: “intrinsic energy of inertia”

Force caused by Potential: Electro-Magnetic Field

- static: localized source charge at rest, only spatial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservative}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right) \cdot \frac{v}{c}\right]}{\text{acceleration}}}_{(3)} \right]$$

Electric Field in Matter

Displacement $\hat{=}$ Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle p_m \rangle \quad (4)$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M}$ = Magnetisation

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. excitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$ χ = elec. susceptibility of dielectricum $\kappa = \frac{\mu - \mu_0}{\mu_0}$ = magn. susceptibility

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant ε_0 = ...

Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions)

Static, kinetec rest charge localisation (conservativ system):

...

Stationary, slow charge movement localisation (conservativ system):

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

Dynamic, fast charge movement (relativistic):

...

Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomgenity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

elec. Potential structure

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

magn. Potential structure

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=const}{=} \quad (9)$$

Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu = \hbar\omega$$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} C U^2 \quad (14)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} L I^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \stackrel{electromagnetic}{=} kc \left(\mathcal{E}_\beta^{-1} \mathcal{E}_\alpha \right)^{-1} = kc \frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} \quad (17)$$

Phase:

$$\varphi = k \left(\vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

Phaseshift:

$$\Delta\varphi = \Delta k \left(\vec{r} \cdot \vec{\Delta k} \right) - \Delta\omega t := \Delta ks - \Delta\omega t = \dots \quad (19)$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\lambda_0}{\lambda} = c \sqrt{\varepsilon \mu} \hat{=} \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = \dots \quad (20)$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielectric medium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$\dots \quad (21)$$

Dielectricity:

$$\varepsilon(\omega) = \dots \quad (22)$$

Permeability:

$$\mu(\omega) = \dots \quad (23)$$

Conductivity:

$$\sigma(\omega) = \dots \quad (24)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (25)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (26)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (27)$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \quad (28)$$
$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (29)$$
$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity and periodic (harmonious oscillation) Movement in VACUUM

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu \lambda = \frac{\lambda}{T}$$

$$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

Refraction index η emerges: $\varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$\vec{p}$ = elec. Dipol

Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|}\right)^2} = 1$$

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with dielectricum, refraction merges, reflection, absorption

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\varphi) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{aligned} \quad (30)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (31)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (32)$$

V_0 = Opposing Potential ($I = 0$)

$\varphi = \angle(E_{kin}, \nu)$ e = electron charge Q = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

Ionisation Energy of chemical element $E_i = h\nu = h \frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$

¹Wirkungsquantum

Energy released (gained), see... Equation 6

- with mass (e.g. electron), Tunnel-Effect: $E_{gain} = E_{pot} - E_i = mc^2 - h\nu_i$
- massless (e.g. photon), Doppler-Effect (frequency shift): $E_{gain} = E_{kin} - E_i = h(\nu_0 - \nu_i) = \Delta\omega\hbar$

Electromagnetic Energy Exchange in Vacuum

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \quad (33)$$

$\varrho = \sigma\delta = \varepsilon_0 \nabla \cdot \mathcal{E}$ = charge density in interior

σ = charge density at border (skin)

δ = Dirac delta-function

$j = \sigma(\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t}\mathcal{D} = \frac{1}{\mu_0} \nabla \times \mathcal{B} - \varepsilon_0 \frac{\partial}{\partial t}\mathcal{E} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\bar{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}(w_e + w_m) + \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt}(LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (34)$$

Energy Balance

Electro-Magnetic Energy:

$$E = E_{kin} + E_{pot} = \sqrt{(pc)^2 + (mc^2)^2} \stackrel{\text{vacuum}}{=} pc = \hbar\omega = \text{chargeenergy} + \text{fieldenergy} + \text{transportenergy} \quad (35)$$

Charge Energy:

$$\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} = - \left(\underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}} \right) = \mathcal{E} \cdot \nabla \times \mathcal{A} - \mathcal{E} \cdot \frac{\partial}{\partial t} \mathcal{D} \quad (36)$$

Field Energy:

$$\frac{\partial}{\partial t} w = \frac{\partial}{\partial t} \sum w_i = \frac{\partial}{\partial t} (w_e + w_m) = \frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \stackrel{\text{vacuum}}{=} \frac{\partial}{\partial t} \left(\frac{1}{2} \frac{1}{\mu_0} (\mathcal{A} \cdot \nabla \times \mathcal{B} - \mathcal{B}^2) \right) \quad (37)$$

Transfer Energy:

$$\nabla \cdot \mathcal{S} \stackrel{\text{matter}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{H}) \stackrel{\text{vacuum}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{B}) \stackrel{\text{alternativ in vacuum}}{=} \nabla \cdot \left(\frac{1}{2} \frac{1}{\mu_0} \frac{\partial}{\partial t} (\mathcal{A} \times \mathcal{B}) \right) = \dots \quad (38)$$

Local Energy Conservation:

$$\text{Force } F = q(\mathcal{E} + v \times \mathcal{B})$$

$$\text{Work } W = \Delta E = Fr = pv = \text{const.}$$

$$\text{Power } \frac{d}{dt} W = 0$$

$$\overbrace{\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}}}_{= \text{const.}} \stackrel{\text{alternatives}}{=} j \cdot \mathcal{E} + \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{A} \cdot \nabla \times \mathcal{B} \right) + \frac{1}{\mu_0} \nabla \cdot \left(\mathcal{E} \times \mathcal{B} + \frac{1}{2} \frac{\partial}{\partial t} \right) \quad (39)$$

conditions for modeling alternatives (Eigenvalue solution):

- $w = \sum w_i : \frac{\partial}{\partial t} w = -\nabla \cdot \mathcal{S}_w$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$

Energy Density Flux: Poynting-Vector

$$\mathcal{S} \stackrel{vacuum}{=} \mathcal{E} \times \mathcal{B} \stackrel{matter}{=} \mathcal{E} \times \mathcal{H}$$

Far Field: Power = $\frac{\Delta E}{t} \hat{=} \frac{d}{dt} E = 0 \Rightarrow \text{Energy} = \text{const.} \Rightarrow \text{Energy Conservation}$

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) = 0$$

= const.

Near Field (local): Power = Energytransport

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) \stackrel{local}{=} \underbrace{- \int_{\mathbb{R}^2} \mathbf{n} \cdot \mathcal{S} \, \underbrace{df}_{\text{"skin"}}}}_{\text{transported energy density}}$$

Local Force Balance

Electro-Magnetic Force:

$$\dot{p} = F = q(\mathcal{E} + v \times \mathcal{B}) = \text{chargeforce} + \text{fieldforce} + \text{stresstension} \quad (40)$$

Charge Force:

$$\dots = \rho \mathcal{E} + j \times \mathcal{B} \quad (41)$$

Field Force:

$$\dots = \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S} \quad (42)$$

Tension Force:

$$\nabla \cdot \mathcal{T} = \varepsilon_0 (\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E} \nabla \cdot \mathcal{E}) + \frac{1}{\mu_0} (\mathcal{B} \nabla \times \mathcal{B} - \mathcal{B} \nabla \cdot \mathcal{B}) \quad (43)$$

Local Momentum Conservation:

$$\text{Force } F = q(\mathcal{E} + v \times \mathcal{B}) = \frac{d}{dt} p = 0$$

$$\text{Momentum } p = \int F = \frac{E}{c} = \text{const.}$$

$$\overbrace{\underbrace{\rho \mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force}} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}}}^{\dots} = 0 \quad (44)$$

Tension Force: Stress-Tensor

Stress Tensor quantifies tension $\hat{=}$ Forces = Pressure - Traction $\hat{=}$ “torsion tension” at angular momentum

$$\mathcal{T} \hat{=} \mathcal{T}_{ik} = \delta_{ik} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \right) - \varepsilon_0 \mathcal{E}_i \mathcal{E}_k - \frac{1}{\mu_0} \mathcal{B}_i \mathcal{B}_k \quad (45)$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static:
 $\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T} d^3r = \int_{\infty} \mathcal{T} df = 0$

Far Field: Force = 0 $\Leftrightarrow$ Momentum = const. $\Rightarrow$ Momentum Conservation

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right]}_{=p=const.} = 0$$

Near Field (local): Force = Tension

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right]}_{=p \neq const., \exists -\mathcal{T} \text{ local tension}} \stackrel{local}{=} - \underbrace{\int_{\mathbb{R}^2} \mathbf{n} \cdot \mathcal{T} df}_{\text{stress tension appears "skin"}}$$

Momentum Balance

Momentum Conservation and Local Tension: linear momentum conservation if $p = const.$, otherwise “tension” occurs:

$$p = \int F = \int \frac{d}{dt} p = \sqrt{\frac{1}{c^2} (E^2 - (mc)^2)} \stackrel{vacuum}{=} \frac{E}{c} = \frac{2\pi}{c^2} h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p} \hat{=} \int_{\mathbb{R}^3} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] = 0$$

= p momentum conserved if p = const. (46)

$p \neq \text{const.}$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists -\mathcal{T}$ as tension compensation flux:

$$F = \dot{p} \hat{=} \int_{\text{local}} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{\text{local}} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] \stackrel{\text{local}}{=} - \int_{\mathbb{R}^2} \mathbf{n} \cdot \mathcal{T} \underbrace{d_j}_{\text{"skin"}}$$

= $p \neq \text{const.}$, $\exists -\mathcal{T}$ local tension (47)

$$\text{Momentum } p = \int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$$

$$\text{photon Spin} = \int_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=} \dots \quad (48)$$

Abbreviations:

w_i = energy densities (electric w_e , magnetic w_m , gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectores $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

$\mathbf{n}$ = surface normal vector (orthogonal on skin)

$\Delta \cdot := \text{div grad} = \text{Laplacian}$

J = "ignition of inhomogeneity, cause"

j = Current density

ϱ = Charge density

V = Potential difference

I = Current

R = Resistance

L = Inductance

C = Capacitance

Q = charge (ion)

$\Psi = LI$ = magn. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} =$ skin depth (surface) of conductor

$r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}\rho$ = radius (extension) of conductor

$\rho = kr$ = “radius of curvature k ”

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)

Electromagnetic Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Vacuum

. - Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast) - Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Localization and Fields of Energy Transport

Localization of energy density w and of its energy transport $\mathcal{S}_w$ is where field force has strongest density. Fields for Forces, Energy, Momentum, Matter:

- Electromagnetic:

$$\left. \begin{array}{l} \text{Electric: } \mathcal{E}, \mathcal{D}, \mathcal{P}, \Phi_e \\ \text{Magnetic: } \mathcal{B}, \mathcal{H}, \mathcal{M}, \mathcal{A}, \Phi_m \end{array} \right\} \mathcal{S}, \mathcal{T}$$

- Thermal (skalar): T
- Gravitational: G

Electromagnetic Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Vacuum

. - Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast) - Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Localization and Fields of Energy Transport

Localization of energy density w and of its energy transport $\mathcal{S}_w$ is where field force has strongest density. Fields for Forces, Energy, Momentum, Matter:

- Electromagnetic:

$$\left. \begin{array}{l} \text{Electric: } \mathcal{E}, \mathcal{D}, \mathcal{P}, \Phi_e \\ \text{Magnetic: } \mathcal{B}, \mathcal{H}, \mathcal{M}, \mathcal{A}, \Phi_m \end{array} \right\} \mathcal{S}, \mathcal{T}$$

- Thermal (skalar): T
- Gravitational: G

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta kv = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \stackrel{\text{electromagnetic}}{=} \dots = \sqrt{\frac{1}{LC}} \stackrel{QM}{=} \frac{1}{2} \frac{p^2}{m \hbar} \quad (49)$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v} \frac{1}{\lambda} = \frac{\omega}{k} \frac{1}{\lambda} = \frac{\omega}{2\pi} \quad (50)$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{h}\omega\mathcal{T} + \nu_0 \quad (51)$$

$$\begin{aligned} e &= \text{electron charge} & V_0 &= \text{Opposing Potential } (I = 0) \\ \text{Energy Level: } i &= [1, n] \in \mathbb{N} \end{aligned} \quad (52)$$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \quad (53)$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurrence of an event/state by space distance and time periodicity as well as movement speed and reach length, $v^2 = v_{ph}v_{gr}$, $\omega t = 2\pi\nu t = \lambda \neq 0$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \quad (54)$$

Rest state of source charge

Statics localization of charge. Only spacial demependency, no time dependency.

Energy transport ... Information transmission ...

$$v = 0 \quad \dot{v} = 0$$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v = \text{const.} \ll c \quad \dot{v} = 0$$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \approx c \quad \dot{v} = 0$$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \neq 0 \quad \dot{v} \neq 0$$

Movement in Vacuum

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = 1 \quad 2\pi = k\lambda \quad v_{ph} = c \quad \omega = kc$$

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

$$\text{Continuity: } \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Movement in Matter

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \neq 1 \quad v_{ph} = \pm \sqrt{\frac{1}{\varepsilon \mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \quad \omega = kv_{ph} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} =$$

$$kc \frac{\tan \beta}{\tan \alpha} = k \frac{c}{\eta}$$

$$\varepsilon \mu = \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon \mu \omega^2 + i\mu \sigma \omega$$

Near Field

$$r_0 \ll r \ll \lambda \quad kr \ll 2\pi \quad \frac{w_m}{w_e} \ll 1$$

r is small, fast movement

Far Field

$$r \gg \lambda \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1 \quad \frac{w_m}{w_e} = 1$$

r is big, slow movement

High Frequency

$$\omega \gg \frac{2}{\mu \sigma r_0^2}$$

λ is big, ν frequency is small, short time period

Low Frequency

$$\omega \ll \frac{2}{\mu \sigma r_0^2}$$

λ is small, ν frequency is big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global:

Local:

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1} \mathcal{D} = \mathcal{E}^{-1} (\mathcal{P} + \varepsilon_0 \mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1} \mathcal{H} = \mathcal{B}^{-1} \left(\frac{1}{\mu_0} \mathcal{B} - \mathcal{M} \right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1 + \omega^2 \tau^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Conditions (skin, interface, border, surface)

Boundary Conditions

$$\mathbf{n} \cdot [\mathcal{D}] = \sigma$$

$$\mathbf{n} \cdot [\mathcal{B}] = 0$$

$$\mathbf{n} \times [\mathcal{E}] = 0$$

$$\mathbf{n} \times [\mathcal{H}] = J$$

Interior Conditions

e.g. for Superconductivity

$$\mathbf{n} \cdot \mathcal{H} = 0$$

$$\mathbf{n} \times \mathcal{D} = 0$$

Abbreviations

$\mathbf{n}$ = normal vector (orthogonality orientation)

$\mathcal{E}$ = elec. Field (“traction force”?)

$\mathcal{B}$ = magn. Field (“compressive force”?)

$\mathcal{D}$ = elec. Displacement Field (“exitioan”)

$\mathcal{H}$ = magn. Exitation Field (“incentivation”)

σ = electric charge at surface

J = current density at surface

Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentiating situations (conditioning frameworks) for fields and potentials:

Spacetime Embedding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
 - Stationary: slow motion
 - Relativistic: fast motion
 - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
 - Mass (inertia)
 - Charge (conductor, ion)
 - Dielectricum (isolator)
 - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reactio, homogeneity vs. inhomogeneity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε , permeability μ , conductivity σ , frequency ω - characteristics from heat: ... - Properties from gravitation: ...

Operators

$\nabla := grad$ = Gradient (Gefälle, Neigung)

$\nabla \cdot := div$ = Divergence (Quelle)

$\nabla \times := rot$ = Rotation (Wirbelung, Curl)

∇^2 = Lagrange Operator (Krümmung, Beschleunigung)

$\Delta \cdot := div(grad)$ = Laplace Operator

$\Delta :=$ Difference, shift